```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

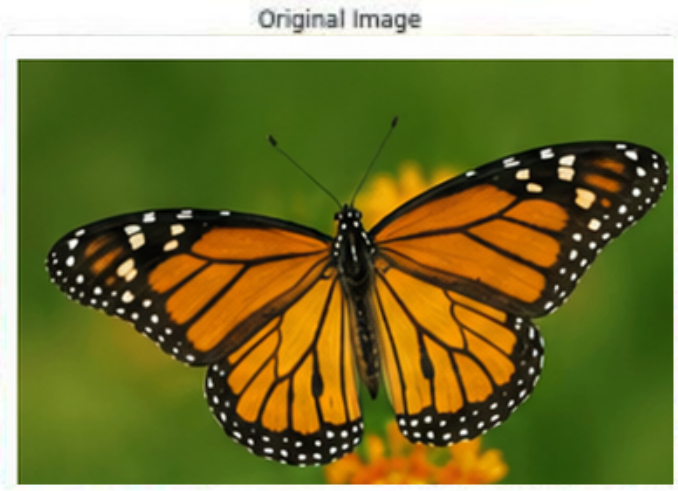
uploaded = files.upload()
img_path = list(uploaded.keys())[0]
img = cv2.imread(img_path)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

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Saving WhatsApp Image 2026-07-23 at 16.43.36.jpeg to WhatsApp Image 2026-07-23 at 16.43.36 (2).jpeg

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image"); plt.axis('off'); plt.show()
```




```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

gaussian = cv2.GaussianBlur(img, (51,51), 0)
plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')
plt.subplot(1,2,2)
plt.imshow(cv2.cvtColor(gaussian, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Blur")
plt.axis('off')
plt.show()
```



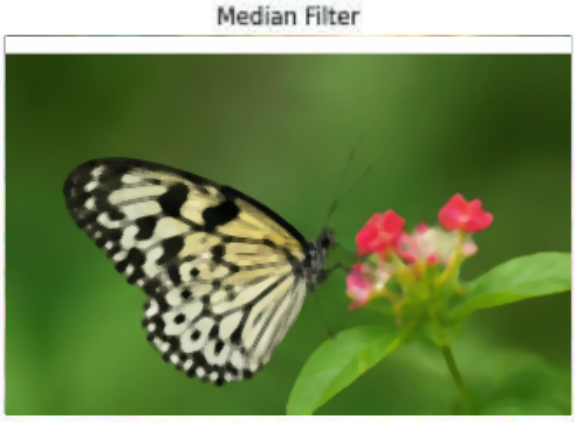



```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

median = cv2.medianBlur(img,31)
plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(cv2.cvtColor(median, cv2.COLOR_BGR2RGB))
plt.title("Median Filter")
plt.axis('off')
plt.show()
```






```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

uploaded = files.upload()
img_path = list(uploaded.keys())[0]
img = cv2.imread(img_path)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

average = cv2.blur(img, (25,25))

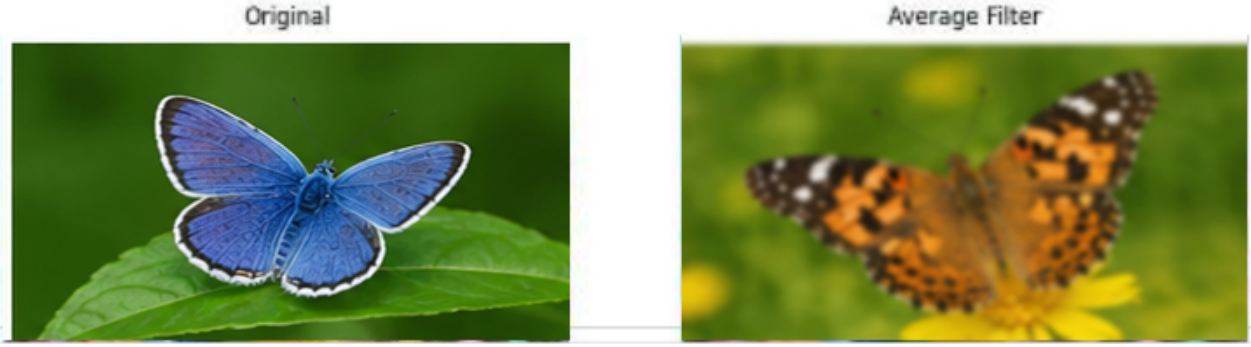
plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(1,2,2)
```

```
plt.imshow(cv2.cvtColor(average, cv2.COLOR_BGR2RGB))
plt.title("Average Filter")
plt.axis('off')
plt.show()
```

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```
Saving WhatsApp Image 2026-07-23 at 16.43.40 (1).jpeg to WhatsApp Image 2026-07-23 at 16.43.40 (1) (1).jpeg
```

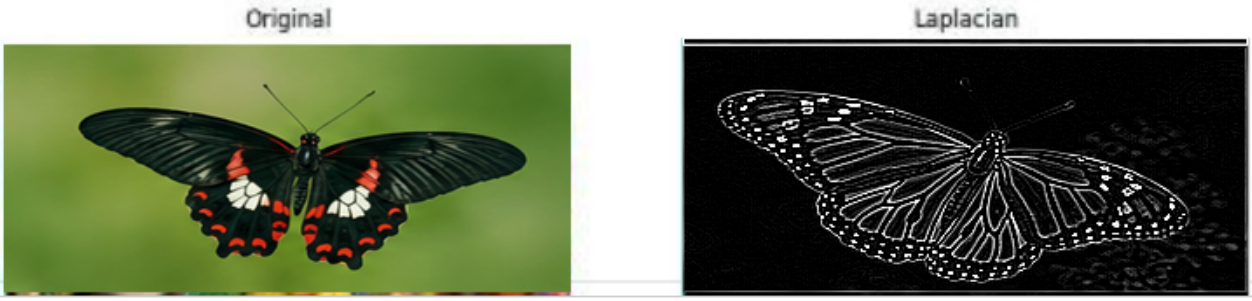



```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(laplacian, cmap='gray')
plt.title("Laplacian")
plt.axis('off')
plt.show()
```




```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow
import matplotlib.pyplot as plt

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)
sobel_combined = cv2.bitwise_or(sobel_x, sobel_y)

plt.figure(figsize=(16,5))
plt.subplot(1,3,1)
plt.imshow(sobel_x, cmap='gray')
plt.title("Sobel X")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(sobel_y, cmap='gray')
plt.title("Sobel Y")
```

```
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(sobel_combined, cmap='gray')
plt.title("Sobel Combined")
plt.axis('off')
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

gaussian = cv2.GaussianBlur(img, (25,25), 0)
median = cv2.medianBlur(img, 15)
average = cv2.blur(img, (25,25))

laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)
sobel_combined = cv2.bitwise_or(sobel_x, sobel_y)

titles = ['Original', 'Gaussian Blur', 'Median Filter', 'Average Filter',
          'Laplacian', 'Sobel X', 'Sobel Y', 'Sobel Combined']
images = [img, gaussian, median, average, laplacian, sobel_x, sobel_y, sobel_combined]

plt.figure(figsize=(16,10))
for i in range(len(images)):
    plt.subplot(2,4,i+1)
    if len(images[i].shape) == 3:
        plt.imshow(cv2.cvtColor(images[i], cv2.COLOR_BGR2RGB))
    else:
        plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')
plt.tight_layout()
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

gaussian = cv2.GaussianBlur(img, (25,25), 0)
median = cv2.medianBlur(img, 15)
average = cv2.blur(img, (25,25))
```

```
laplacian = cv2.Laplacian(gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)
sobel_combined = cv2.bitwise_or(sobel_x, sobel_y)

# Original + saare filtered images ek saath display
titles = ['Original', 'Gaussian Blur', 'Median Filter', 'Average Filter',
          'Laplacian', 'Sobel X', 'Sobel Y', 'Sobel Combined']
images = [img, gaussian, median, average, laplacian, sobel_x, sobel_y, sobel_combined]

plt.figure(figsize=(16,10))
for i in range(len(images)):
    plt.subplot(2,4,i+1)
    if len(images[i].shape) == 3:
        plt.imshow(cv2.cvtColor(images[i], cv2.COLOR_BGR2RGB))
    else:
        plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')
plt.tight_layout()
plt.show()

print("Observations:")
print("- Gaussian Blur: Noise kam karta hai, image ko smooth banata hai, edges thoda soft ho jate hain.")
print("- Median Filter: Salt-and-pepper noise bahut achhe se hatata hai, edges better preserve karta hai.")
print("- Average Filter: Simple smoothing karta hai, lekin edges Gaussian se zyada blur ho jate hain.")
print("- Laplacian Filter: Image ke edges aur fine details ko highlight karta hai (high-pass filter).")
print("- Sobel X/Y: Horizontal aur vertical gradients detect karta hai, edges ki direction pata chalti hai.")
```




```
Observations:
- Gaussian Blur: Noise kam karta hai, image ko smooth banata hai, edges thoda soft ho jate hain.
- Median Filter: Salt-and-pepper noise bahut achhe se hatata hai, edges better preserve karta hai.
- Average Filter: Simple smoothing karta hai, lekin edges Gaussian se zyada blur ho jate hain.
- Laplacian Filter: Image ke edges aur fine details ko highlight karta hai (high-pass filter).
- Sobel X/Y: Horizontal aur vertical gradients detect karta hai, edges ki direction pata chalti hai.
- Sobel Combined: Dono directions ke edges ek saath dikhata hai, complete edge map milta hai.
```